Yichao Cai, PhD Candidate in Computer Science

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Research Profile

I study how learning objectives and supervision signals shape representations: when they identify causal and generalizable latent structure, and when they discard, conflate, or leave it underdetermined. My work combines identifiability theory, latent-variable modeling, population-objective analysis, and representation geometry. By revealing what current objectives can and cannot recover, it develops principles for designing objectives, supervision, and data interventions that produce more reliable and generalizable multimodal and foundation models.

Research Interests

Representation learning; multimodal learning; causal and identifiable representation learning; foundation-model objectives.

Education

PhD Candidate in Computer Science — Sep 2023–Present
Australian Institute for Machine Learning, Adelaide University
Advisor: Prof. Javen Qinfeng Shi
Research: representation learning, causal machine learning, multimodal learning

M.Sc. in Instrument Science and Technology — Sep 2016–Jun 2019
Wuhan University of Technology
Advisor: Prof. Xiao Zhou
Research: image processing, computer vision, and autonomous-driving perception

B.Eng. in Measurement and Control Technology and Instrument — Sep 2012–Jun 2016
Wuhan University of Technology

Peer-reviewed Publications

[1] **Cai, Y.**, Zhang, Z., Liu, Y., and Shi, J. Q. "The Geometric Mechanics of Contrastive Representation Learning: Alignment Potentials, Entropic Dispersion, and Cross-Modal Divergence." International Conference on Machine Learning (**ICML**), 2026.

[2] **Cai, Y.***, Liu, Y.*, Gao, E., Jiang, T., Zhang, Z., van den Hengel, A., and Shi, J. Q. "On the Value of Cross-Modal Misalignment in Multimodal Representation Learning." Advances in Neural Information Processing Systems (**NeurIPS**), 2025. **Spotlight**.
*Equal contribution.

[3] **Cai, Y.**, Liu, Y., Zhang, Z., and Shi, J. Q. "CLAP: Isolating Content from Style Through Contrastive Learning with Augmented Prompts." European Conference on Computer Vision (**ECCV**), 2024.

[4] Liu, Y., Gong, D., **Cai, Y.**, Gao, E., Zhang, Z., Huang, B., Gong, M., van den Hengel, A., and Shi, J. Q. "I Predict Therefore I Am: Is Next Token Prediction Enough to Learn Human-Interpretable Concepts from Data?" International Conference on Learning Representations (**ICLR**), 2026.

[5] Jiang, W., Liu, Y., **Cai, Y.**, Gao, E., Dong, J., Abbasnejad, E., Yao, L., and Shi, J. Q. "What Makes a Representation Good for Single-Cell Perturbation Prediction?" International Conference on Machine Learning (**ICML**), 2026.

[6] Chen, J., Yin, Z., **Cai, Y.**, Liu, Y., Zhang, Z., Gong, D., and Shi, J. Q. "Boundary Embedding Shaping with Adaptive Contrastive Learning for Graph Structural Disentanglement." International Conference on Machine Learning (**ICML**), 2026.

Preprints

[1] **Cai, Y.** and Shi, J. Q. "On the Identifiability of Masked Prediction: Mode Blindness and Mask Schedules." arXiv:2608.01383, 2026.

Invited Talks

Understanding Multimodal Contrastive Learning through Two Lenses — Jul 2026
Online invited talk, Vision & Sensing Lab, Technion
Online invited talk, EML, Helmholtz Munich

Honors & Awards

D. R. Stranks Travelling Fellowship — Jun 2026
NeurIPS Scholar Award — Oct 2025
University of Adelaide Research Scholarship — Sep 2023
Outstanding Graduate, Wuhan University of Technology — Jun 2019

Teaching Experience

Adelaide University
(Formerly the University of Adelaide)

Teaching Assistant, *Programming for Artificial Intelligence* — Semester 2, 2026

Teaching Assistant, *Neural Networks and Deep Learning* — Semester 1, 2026

Guest Lecturer & Head Tutor, *Statistical Machine Learning* — Semester 2, 2025
Delivered a guest lecture on vision-language modeling and coordinated the tutor team.

Teaching Assistant, *Using Machine Learning Tools* — Trimester 2, 2025
Developed automated scripts for efficient and consistent assignment marking.

Teaching Assistant, *Concepts in AI and Machine Learning* — Semester 1, 2025

Academic Service

Conference reviewer: AAAI 2027; NeurIPS 2026; ICML 2026; ICLR 2026.
Journal reviewer: Transactions on Machine Learning Research (TMLR).

Professional Experience

AI Engineer — Jun 2020–Aug 2022
Artificial Intelligence Research Institute, Tellhow Software
- Developed computer-vision and automated-analysis systems for power-equipment inspection and safety monitoring.
- Contributed to algorithm validation, error analysis, system integration, deployment, and technical documentation.

Software Engineer — Jul 2019–Apr 2020
Huawei Technologies
- Developed, tested, and integrated production communication-service components.
- Conducted feature implementation, interface debugging, and production issue diagnosis in a collaborative engineering environment.

Visiting Research

Visiting Student Researcher — May 2018–Sep 2018
California PATH, UC Berkeley
Conducted research on visual-perception algorithms for autonomous driving, including model development and experimental evaluation.

Technical Expertise

Methods: identifiability analysis, latent-variable modeling, population-objective analysis, representation geometry, probing, controlled experimentation, and benchmark design.
Software: Python, PyTorch, C/C++, Linux, and Git.
Languages: Mandarin Chinese (native); English (professional working proficiency).